

A Note on the Syntax and Semantics of Chinese Classifiers^{*}

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Michael Barrie and Tianyu Zhao. 2025. **A Note on the Syntax and Semantics of Chinese Classifiers**. *Studies in Generative Grammar*, 35-2. 253-273. We examine the structure of classifier constructions in Mandarin Chinese in light of Little *et al.*'s (2022) proposal on the semantics of two kinds of classifier constructions and Zhang's (2011, 2013) analysis of classifiers in Chinese. Little *et al.* propose that classifiers either compose first with nouns (classifiers-for-nouns) or compose first with numerals (classifiers-for-numerals). The semantic denotation of numerals is different in both cases. Zhang proposes that Chinese has both structures. Taking both proposals at face value would require us to assume that numerals are listed twice in the lexicon in Chinese (once for each denotation in Little *et al.*'s models). We critically examine Zhang's diagnostics and conclude that Chinese has only the classifier-for-nouns structure. A tentative semantic analysis is provided.

Keywords: *repair, island violation, ellipsis, sluicing, fragment, phase, contrast, LBC*

1. Introduction

The properties of classifiers in Chinese have been extensively studied both within generative frameworks (Cheng & Sybesma 2012; Cheng & Sybesma 2005; Cheng & Sybesma 1998; Jiang *et al.* 2022) and in the descriptive literature (Chao 1968; Peyraube 1991; Allan 1977; Zhang 2007; Li & Thompson 1981; T'sou 1976). There

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have been two major approaches to the syntax and semantics of numeral-classifiers (Krifka 1995; Chierchia 1998; Bale & Coon 2014; Little *et al.* 2022). The first of these is one in which the classifier and the noun form a syntactic constituent and compose semantically. We will call this the classifier-for-nouns model (1a). The other is when the classifier and the numeral form a syntactic constituent and compose semantically (1b). We use the # symbol here for the numeral. We call this the classifier-for-numerals model. Note that the structures in (1) are intended to show constituency only and not headedness. We examine both the syntactic and semantic implications of these two proposals in the context of a variety of classifier constructions in Mandarin Chinese (henceforth simply Chinese).¹

- (1) a. [# [CL N]]
 b. [[# CL] N]

It has been suggested that classifiers in Mandarin Chinese exhibit both structures in (1) above (Zhang 2011; Li 2014). Furthermore, Little *et al.* propose an attractive theory of the semantics of classifiers that distinguishes between the two models in (1). They adopt a uniform denotation for the semantics of nouns; however, they crucially adopt a different denotation for numerals in the two models. Thus, for example, the denotation for a numeral such as three in a classifier-for-nouns language is different from the denotation of the same numeral in a classifier-for-numerals language.

As mentioned above, it has been argued that Chinese has both types of classifier constructions. Taking Zhang's analysis and Little *et al.*'s model at face value, we would be forced to assume that Chinese has two lexical entries for every numeral, a rather unparsimonious analysis.

The remainder of this paper is structured as follows. Section 2 presents the background for this study, presenting the core aspects of Little *et al.*'s proposal and introducing the various kinds of classifiers in Mandarin Chinese. Section 3 critically reviews the previous research on Chinese classifiers and introduces Little *et al.*'s (2022) proposal for two models of classifier constructions. Section 4 discusses Zhang's (2011, 2013) diagnostics for the syntactic structure of classifier constructions in Chinese and some problems therein. It also discusses diagnostics

¹ Li (2014) concludes that #-CL-N constructions in Chinese uniformly have the classifier-for-nouns model as we argue here. She also discusses predicative uses of numerals such as *san bang de xigua* (three pound DE watermelon - 'a three-pound watermelon'), in which she argues that the numeral and the classifier form a constituent. Such forms have a significantly different semantic structure than the forms under consideration here, so we leave them to future research.

from Little *et al.* designed to choose between the two models for classifier constructions. We conclude that classifier constructions in Chinese are uniformly of the classifier-for-nouns type. Section 5 is a discussion and presents a tentative semantic analysis for individual classifiers and container classifiers. Finally, section 6 is a brief conclusion.

2. Background

2.1. Two Kinds of Classifier Construction

Two syntactic structures have been entertained for classifier constructions, both in Chinese languages and in other languages. As mentioned, we refer to these as the “classifiers-for-nouns” model, (2a), and the “classifiers-for-numerals” model, (2b).

- (2) a. [CIP # [CI' CI [NP N]]]
 b. [XP [YP # CI] [X' X [NP N]]]

The crucial difference between these two structures is the following. In the classifiers-for-nouns model, the classifier and the noun form a constituent to the exclusion of the numeral (#). In the classifier-for-numerals structure, the classifier and the numeral form a constituent to the exclusion of the noun. The precise details vary from one analysis to the next, but the constituency facts as just explained hold across different analyses.

The semantics of classifiers has given rise to the same dichotomy. The most detailed semantic analysis of these two structures is found in Little *et al.* (2022). In both structures, the denotation of the noun is the same, namely an atomic join semi-lattice in the sense of Link (1983). For simplicity, the definition below assumes that there are three dogs in the world.

- (3) $\llbracket \text{DOGS} \rrbracket = \lambda x. [\text{DOGS}(x)] = \{a, b, c, a+b, a+c, b+c, a+b+c\}$

The denotation in (3) builds the denotation of the plural noun from the atomic set DOG with the * operator familiar from Link. The set DOGS, then, is the denotation of a predicate that is syntactically more complex than that of which the set DOG is the denotation. It would typically be assumed to be a NumP in English. Little *et al.* do not discuss this issue in detail in their paper; however, we will touch on this issue below. For now, we simply introduce the two

semantic derivations.

In the classifiers-for-nouns approach, the classifier composes with the noun to give rise to an atomic set.

- (4) a. $\llbracket \text{CL} \rrbracket = \lambda P \lambda x. [P(x) = 1 \wedge [\nexists y | P(y) \wedge y \leq x]]$
 b. $\llbracket \text{CL} + \text{DOGS} \rrbracket = \lambda x. [\text{DOGS}(x) \wedge [\nexists y | \text{DOGS}(y) \wedge y \leq x]] = \{a, b, c\}$
 c. $\llbracket \text{two} \rrbracket = \lambda P \lambda x. [\exists S [\Pi(S)(x) \ \& \ |S| = 2 \ \& \ \forall s \in S [P(s)]]]$

In the classifiers-for-numerals approach, the classifier composes with the numeral to make it suitable for counting. In this theory, numerals lack a measure function, which is provided by the classifier. Little *et al.* use μ to represent a variable over measure functions.

- (5) a. $\llbracket \text{CL} \rrbracket = \mu \#$
 b. $\llbracket \text{two} \rrbracket = \lambda m \langle e, n \rangle \lambda P \lambda x. [P(x) \ \& \ m(x) = 2]$
 c. $\llbracket \text{two} + \text{CL} \rrbracket = \lambda P \lambda x. [P(x) \ \& \ \mu(x) = 2]$

These two theories of classifier constructions make clear and testable predictions, some of which we run through here. What Little *et al.* argue is that some languages have the classifiers-for-nouns model and some have the classifiers-for-numerals model. A further prediction seems to be that it would be surprising for a language to have both types. This is because the denotations of the numerals are different between the two models. Such a language would have to have two lexical entries for every numeral, or at least every morphologically unique numeral². Such a language might be expected to have two morphologically distinct forms for every numeral.

2.2. Mandarin Chinese Classifiers

A number of kinds of classifiers have been identified in Mandarin Chinese. We review their basic properties here. The classification presented here is based largely on Zhang (2011) and Jiang *et al.* (2022). The reader is referred to these two works for further details. First, we consider individual and individuating classifiers. Individual classifiers appear with ontologically count nouns and are a virtually universally accepted category of classifiers. Consider the following

² Complex numerals, such as 245, are composed semantically from their components. See Ionic and Matushansky (2018) for the composition of complex numerals.

representative examples³.

- (6) Shuzhi shang luo-zhe qi zhi xiao-niao.
 branch on perch-DUR seven CL small-bird
 'Seven small birds are perched on the branch.'

Individuating classifiers parcel out mass substances into countable units. These are discussed by Zhang, but not by Jiang *et al.*

- (7) Ta wang guo-li dao-le san di you.
 3 toward pot-into poured-PRFV three CL oil
 'He poured three drops of oil into the pot.'

Next we consider standard measures and container measures, discussed both by Zhang and Jiang *et al.* Standard measure classifiers measure out a substance into a quantity.

- (8) Ta zai shangdian mai-le wu mi huabu.
 3 in store buy-PRFV five meters flower-cloth
 'He bought five metres of flower-cloth in the store.'

Container measure classifiers measure out mass or count substances into countable containers.

- (9) Weile shenghuo, ta yongdiao-le shi he huochai.
 in.order.to make.fire, 3 use-PRFV ten boxes match
 'In order to make fire, he used 10 boxes of matches.'

Next is collective classifiers (called 'group' by Jiang *et al.*) and partitive classifiers (not discussed by Jiang *et al.*). Collective classifiers count out discrete quantities of objects.

- (10) Bangongzhuo shang bai-zhe ta-de san fen wenjian.
 office-desk on put-DUR 3-DE three CL document
 Three piles of documents are put on his office desk.

³ Abbreviations include CL - classifier, DE - 'de', an attributive marker in Mandarin Chinese, DEM - demonstrative, DUR - durative, PRFV - perfective, PROG - progressive

Finally, partitive classifiers specify portions of a whole.

- (11) Ta yikouqi chi-le wu kuai paigu.
 He in.one.go eat-PRFV five CL spare.ribs
 'He ate five spare ribs in one go.'

It is these different kinds of classifiers that Zhang analyzes as falling into two main types as discussed in the next section. In short, she proposes that the individual and individuating classifiers have the classifier-for-nouns structure, while the others have the classifier-for-numerals structure. We move on to her analysis now and discuss various shortcomings before presenting our analysis.

3. Previous Research on Chinese Classifiers

3.1. Two Structures for Chinese

Although both the classifier-for-nouns and the classifier-for-numerals structures have been proposed for Chinese in general, the classifier-for-nouns structure is by far more common (Chierchia 1998; Cheng & Sybesma 2012; Li 1999). The classifier-for-numerals structure was originally proposed for Chinese by Krifka (1995). This model has since been advanced by Bale and Coon (2014), although not for Chinese *per se*. This structure is by far less commonly pursued for Chinese, but Yang (2001), Tang (1990), and Her and Tsai (2020) do present in depth analyses. A third option is that both types of structures are instantiated in Chinese. Proponents of this approach include Zhang (2011; 2013), Jin (2020), Li (2013), and Jiang (2017).

It is the third option that is of interest to us here. What these analyses have in common is that the classifier constructions used for counting tend to have the classifier-for-numeral structure and the classifier constructions used for measuring tend to have the classifier-for-numeral structure. We concentrate on Zhang's (2013) analysis here, so we comment on it in detail. Zhang specifically argues that the individual and individuating classifiers have the classifiers-for-nouns structure while the other classifiers have the classifiers-for-numerals structure. Her diagnostics will be discussed in section 4.

3.2. Semantics of Numeral Classifiers: Little *et al.*

Little *et al.* (2022) propose a comprehensive framework for understanding the syntactic and semantic roles of numeral classifiers across languages, using empirical data from two typologically distinct languages, Ch'ol (Mayan) and Shan (Kra-Dai). Their proposals identify two fundamentally different types of numeral classifier systems, which are tied to distinct syntactic configurations and semantic contributions. These two systems are categorized as classifiers-for-numerals and classifiers-for-nouns. The discussion below gives the general properties of classifiers in Ch'ol and Shan. The reader is urged to consult Little *et al.* (2022) for an in depth discussion of the data.

In Ch'ol, classifiers primarily serve to provide a measure function for numerals, enabling numerals to combine with nouns. The numeral and classifier form a tightly bound syntactic unit, and classifiers appear obligatorily in counting contexts, making Ch'ol a clear example of a classifiers-for-numerals language. Numerals and classifiers are syntactically adjacent and form a constituent, excluding the noun. Classifiers must appear in all numeral constructions, even without overt nouns, such as in counting (*How many people are there? Let's see. One, two, three...*) or mathematical expressions (*Two plus two equals four*). Numerals occupy a high position in the nominal structure, with classifiers serving as their complements. Semantically, Ch'ol numerals depend on classifiers, and the absence of a classifier renders numeral constructions ungrammatical.

In Shan, classifiers fulfill a distinct role by atomizing the noun's referents into discrete, countable units rather than mediating between numerals and nouns. This allows classifiers to pair with nouns even in the absence of numerals, suggesting that Shan is a clear example of a classifiers-for-nouns language. Numerals and classifiers are not tightly bound; instead, classifiers appear adjacent to nouns while numerals occupy separate positions in nominal phrases. Classifiers in Shan demonstrate flexibility by appearing with quantifiers, demonstratives, and relative clauses, even without numerals. The choice of classifier depends on the semantic properties of the noun, making classifiers integral to the noun's syntactic structure. Certain nouns inherently denote atomic sets, enabling them to appear without classifiers in specific contexts.

Based on these observations, Little *et al.* propose a cross-linguistic distinction in numeral classifier constructions. (1) Classifiers-for-numerals: these classifiers mediate between numerals and nouns, providing a measure function to numerals. This structure is exemplified by Ch'ol, where numeral-classifier constructions are tightly bound and classifiers are obligatory in all numeral contexts. (2) Classifiers-for-nouns: these classifiers modify the semantics of nouns, atomizing their referents and

preparing them for numerical or other types of modification. Shan exemplifies this structure, where classifiers can occur independently of numerals and are more closely tied to the noun. The following section discusses various diagnostics for choosing between these two structures and applies them to Chinese.

4. Diagnosing the Constituency of #-CL-N

4.1. Zhang's Diagnostics

We consider the diagnostics Zhang (2011, 2013) uses to distinguish between the two syntactic structures. First, Zhang argues that the appearance of two incompatible adjectives in one and the same nominal requires two different structures for classifiers. Note the following contrast.

- (12) a. *dada de yi wan xiao yingtao.* [container]
 big DE one CL small cherry
 ‘a big bowl of small cherries’
- b. * *dada de yi li xiao yingtao.* [individual]
 big DE one CL small cherry
 (‘#a big (unit of a) small cherry’)

Zhang shows that incompatible adjectives are possible only with the classifiers of the collective and container type. We have shown examples just of the container and individual type of classifier. See Zhang for the full set of data. In Zhang’s opinion, the two different modification domains belong to two different constituents. The first modifier, *dada de* (‘big’), modifies the first constituent *wan* (‘CL.bowl’). The second modifier, *xiao* (‘small’), modifies the head noun *yingtao* (‘cherry’). Therefore, these kinds of classifiers have the classifier-for-numeral structure. The two structures she proposes are shown in (13). Zhang argues that the lack of c-command between the two incompatible adjectives allows the structure to survive. She correspondingly argues that the infelicity of (12b) arises from the c-command relation between these two adjectives, (13b). Specifically, the noun *cherry* is c-commanded by two contradictory adjectives (*big* and *small*).

- (13) a. [[big DE one CL.bowl] small cherry]
 b. [big DE [one [CL small cherry]]]

We note various problems with this diagnostic. In (12a), the modifier does not modify the classifier *wan* ('bowl'). Rather, it modifies the whole DP. Namely, it refers to a large quantity of cherries. The bowl itself is not necessarily big. Crucially, this expression cannot be used to describe a large bowl that contains a small amount of cherries. Thus, we suggest that 'big' scopes over 'bowl of cherries' and that 'small' scopes over 'cherries'. Thus, there is no need to invoke c-command or lack thereof to explain the facts in (12). The following example is even more damaging to Zhang's analysis.

- (14) *lǜlǜ de yī wǎn yíngtáo*
 green DE one CL.bowl cherry
 'a bowl of green cherries' (lit: a green bowl of cherries)

In this example, crucially, the cherries are green, not the bowl. It is unclear how the adjective phrase *lǜlǜ de* ('green') could modify the head noun *yíngtáo* ('cherry') under Zhang's analysis as the adjective does not c-command the noun in her analysis.

A reviewer mentions the following example, however.

- (15) *mǎnmǎn de yī wǎn yíngtáo*
 full DE one CL.bowl cherry
 'a full bowl of cherries'

In (15), the adjective clearly does not modify the cherries. We suggest an analysis similar to what we propose above. Namely the adjective modifies "bowl of cherries." If the adjective modified only "bowl," it would be compatible with a scenario in which a bowl is full of some other substance, say water, and contains a few cherries being rinsed in the water. The bowl is full, and the salient contents of the bowl are the cherries. Nevertheless, the expression in (15) cannot be used to describe this situation. It is clear, then, that the adjectives in (12a), (14), and (15) do not modify the classifier. Finally, to explain the contrast in (12), we propose that (12a) has two different concepts that can be modified: "cherry" and "bowl of cherries." In (12b) only "cherry" is a meaningful concept that can be modified, leading to the contradiction.

In another diagnostic, Zhang observes that some modifiers can appear to the left of either the classifier or the head noun with no change in meaning, as shown in the following example.

- (16) yi chang tiao xianglian = yi tiao chang xianglian
 one long CL necklace one CL long necklace
 ‘one long necklace’ [individual]

Such identity in meaning is possible only with individual classifiers. Zhang notes that individuating classifiers are not testable because the relevant modifiers are not possible. With all other types of classifiers a difference in meaning obtains, as shown in the following example.

- (17) yi chang xiang xianglian ≠ yi xiang chang xianglian
 one long CL.box necklace one CL.box long necklace
 ‘one long box of necklaces’ ‘one box of long necklaces’ [container]

We note that container classifiers indicate the size of the amount measured out while individual classifiers merely indicate a canonical unit of a count noun. As such, modifying the size of the counter (here, *xiang* ‘box’) gives rise to a difference in the amount of what is being enumerated, in this case, the amount of necklaces. We propose that container classifiers and the like have a denotation that allows them to undergo predicate modification with an adjective, while individual classifiers do not. In (16), then, the classifier *tiao* has nothing that can be modified by the adjective. Thus, regardless of the position of the adjective, it can modify only the head noun, *xianglian* (‘necklace’).

Zhang notes the semantic selection between the noun and individual and individuating classifiers as evidence for the classifiers-for-nouns structure. Likewise, the absence of such selection, she argues, is evidence for the classifiers-for-numerals structure. The correlation between the noun and the individual classifier, however, is found in languages with strong evidence for the classifiers-for-numerals structure such as Korean and Japanese (see below). Such examples are well known, but we include a few here for illustration for Mandarin Chinese, (18) and Korean (19)⁴.

- (18) a. san pi ma/*zhu b. san zhan deng/*lazhu
 three CL horse/pig three CL lamp/candle
 ‘three horses (*pigs)’ ‘three lamps (*candles)’

⁴ Note that (19a) is grammatical on the reading of “a barrel of cucumbers.” Crucially, the classifiers cannot be used to count individual cucumbers.

- (19) a. swubak/*oi han thong
 watermelon/*cucumber one CL
 ‘one watermelon (*cucumber)’
 b. kulim/*phosuthe han cem
 painting/poster one CL
 ‘one painting (*poster)’

Although there is no clear consensus on the structure of numeral classifiers in Korean (and Japanese), there is some evidence to suggest that the numeral and the classifier have the classifier-for-numerals structure (Kang 2016; Barrie *et al.* 2022; Kim 2005). If so, then the point Zhang makes regarding the selectional restrictions in (18) does not hold water.

Given that the facts in Korean are not conclusive, we also mention Little *et al.*'s (2022) discussion of Ch'ol, where they present clear evidence for the classifier-for-nouns structure for Ch'ol. They note that Ch'ol, like Chinese and Korean, involves semantic selectional restrictions between classifiers and nouns. They note that Ch'ol has at least 180 classifiers, which correlate with the noun, such as *-kojty* for animals, *-tyikil* for humans, *-k'ej* for flat round objects, just to name three. Thus, this variation cannot be taken as evidence for deciding between the two models.

There are various ways to analyze the facts in (18) and (19). Rather than selection, the choice of classifier could simply be allomorphy, which does not require sisterhood. Also, Nomoto (2013), following McCready (2009), argues that the different forms of the classifier arise by pragmatic implicature rather than by selection. We realize there is a huge literature on Korean and Japanese numeral classifiers that cannot be adequately addressed here. Our intention is merely to show that Zhang's claim that the correlation between individual classifiers and nouns does not necessarily lead to the conclusion that they form a constituent, nor that the lack of such a correlation with other classifiers leads to the conclusion that they're not.

We have seen that Zhang's diagnostics do not hold up to scrutiny. There are additional facts about the syntax of Chinese numeral classifiers that might support a classifiers-for-numerals approach, however. One is the observation that the numeral-classifier sequence is rarely if ever interrupted⁵. Consider the following data.

⁵ Thanks to a reviewer for pointing this out.

- (20) a. changchang de yi zhi bi
 long DE one CL pen
 'a long pen'
- b. yi zhi changchang de bi
 one CL long DE pen
 'a long pen'
- c. * yi changchang de zhi bi
 one long DE CL pen
 'a long pen'

Recall from (16) and (17) above that the numeral and classifier can be interrupted by an adjective (see Tang 1990 for discussion), suggesting that the numeral and classifier do not form a constituent. Admittedly, however, the numeral and classifier can be interrupted only by an adjective (*da* 'big') but not by an adjective phrase (*hen da* 'very big'). We note, however, that while the presence of material between the numeral and the classifier offers clear evidence for the classifier-for-noun analysis, lack of such material does not offer clear evidence for the classifier-for-numeral analysis. If the numeral appears in the specifier of CIP and the classifier appears in the head of CIP, for example, there will be very little opportunity for phrasal material to intervene. As a reviewer notes, however, the existence of an adjective between the numeral and the classifier is difficult to reconcile with this proposed structure. We leave this problem to future work.

Finally, it has been noted that the numeral and classifier can appear without a noun, suggesting that they may form a constituent. Here is an example.

- (21) ni chi le duoshao ge pingguo?
 you eat PRFV how.many CL apple
 'How many apples did you eat?'
- (22) (wo chi le) liang ge.
 (I eat PRFV) two CL
 '(I ate) two.'

We suggest that this is the result of *nP* deletion and actually does not diagnose between the two models for classifiers. Specifically, the facts above are compatible with the following structure with *nP* deletion as stated.

(23) [CIP # [CI' CI [nP ...]]]

In the next section, we apply Little *et al.*'s diagnostics to Mandarin Chinese.

4.2. Little *et al.*'s Diagnostics

In this section we cover Little *et al.*'s diagnostics for the syntactic constituency of numerals, classifiers and nouns. First, they argue that the possibility of a classifier and noun to appear together independently of a numeral is evidence for the classifier-for-nouns structure. Consider the following examples. Observe that the classifier and noun can appear with a demonstrative, but no numeral is present. Cantonese additionally shows that the classifier and noun can appear with a possessor.

(24) a. zhe ge pingguo b. zhe wan tang
 DEM CL apple DEM CL.bowl soup
 'this apple' 'that bowl of soup'

(25) ngo go pinggwo [Cantonese]
 1 CL apple
 'my apple.'

It has been argued that examples such as the ones above result from deletion of the numeral *yi* ('one') (see Zhang, 2013 for discussion). Under Little *et al.*'s proposal, however, the denotation of a classifier plus noun is the set of atomic entities as in (4b). Thus, once the expression is either determined by a demonstrative or a choice function, or is existentially quantified, only a singular interpretation is possible. Thus, there is no need to assume that a numeral is present in the derivation only to be elided later.

The other diagnostic Little *et al.* propose is the idiosyncratic behaviour between the classifier and either the noun or the numeral. As discussed above, the classifier-for-nouns structure predicts the possibility of certain nouns that don't require a classifier. Recall that certain classifiers are derived from nouns—the massifiers in the sense of Cheng & Sybesma. Consider the following data.

(26) a. yi ge wan b. yi wan tang
 one CL bowl one CL.bowl soup
 'one bowl' (physical object) 'one bowl of soup'

The data above show that, regardless of classifier type, numerals can be conjoined, but the numeral+classifier cannot be conjoined, thus favouring the classifier-for-nouns model.

Taken together the data in this section largely favour the classifier-for-nouns model for all types of classifiers in Chinese. We move on to the discussion of these results in the next section.

5. Discussion

Little *et al.* (2022) present an attractive analysis of classifiers in which they argue that classifier languages have either a classifier-for-nouns structure, or a classifier-for-numerals structure. Inherent in their analysis is a difference in the semantic composition of the numeral, noun, and classifier. Crucially, the denotations of the numerals differ in the two kinds of languages. They tacitly assume in their discussion that languages have one or the other structure. Zhang (2011, 2013), on the other hand, has argued that Chinese has both types of structures. If both Little *et al.* and Zhang are correct, it would mean that Chinese has two lexical entries for every numeral—a far from parsimonious conclusion.

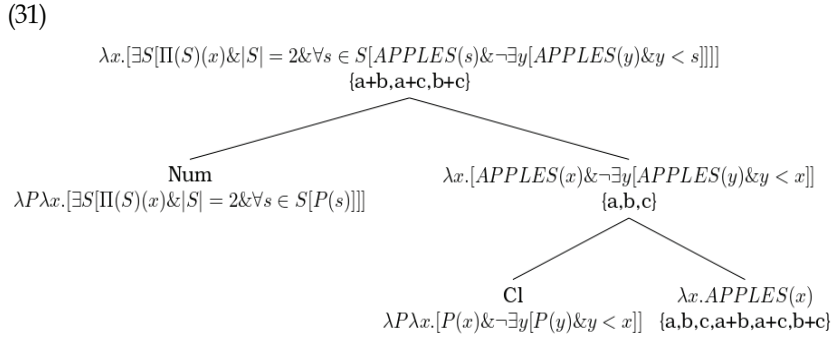
In light of the situation above, we have argued that Zhang's conclusion that Chinese has both structures is not warranted. To recapitulate the results, we have suggested that the facts regarding modifiers can be captured with the classifiers-for-nouns model. Further, it is now well established that at least some classifier languages have the classifiers-for-numerals model. Nevertheless, it is a common property among many classifier languages for the classifier and the noun to correlate (zhi for certain animals in Chinese, kwen for books and printed material in Korean). Furthermore, we presented various diagnostics from Little *et al.* that converged on a classifiers-for-nouns structure for Chinese.

This section makes some suggestions as to how we can implement the semantic analysis in Little *et al.* (2022) to the Chinese data. For succinctness, we deal only with individual and container classifiers, as these were the clearest cases for Zhang's proposal. Starting with the individual classifier, nominals with this classifier have the same derivation as in Little *et al.* Consider the following example.

- (30) a. san ge pingguo
 three CL apple
 'three apples'

- b. $[[pingguo]] = \lambda x.APPLES(x) = \{a, b, c, d, a+b, a+c, a+d, b+c, \dots\}$
 $[[ge]] = \lambda P\lambda x.[P(x) \ \& \ \neg \exists y[P(y) \ \& \ y < x]]$
 $[[san]] = \lambda P\lambda x.[\exists S[\Pi(S)(x) \ \& \ |S| = 3 \ \& \ \forall s \in S[P(s)]]]$

This gives us the structure below.



Deal (2017) proposes that bare nouns universally are of one of the following three types.

- (32) a. quantized nouns - typically count nouns with a number distinction as in English
 b. atomic join-semilattices - typically bare nouns in Chinese, as in (29b)
 c. non-atomic join-semilattices - typically mass nouns

Deal further argues that count and mass nouns have the following denotations. Note that Deal did not explicitly consider count nouns in Chinese, but following her analysis a bare noun in Chinese is cumulative, so is an atomic join-semilattice, not a quantized noun as in English.

- (33) a. $[[cat]]$ = the set of all cat-atoms and their joins
 b. $[[soup]]$ = the set of all portions of soup

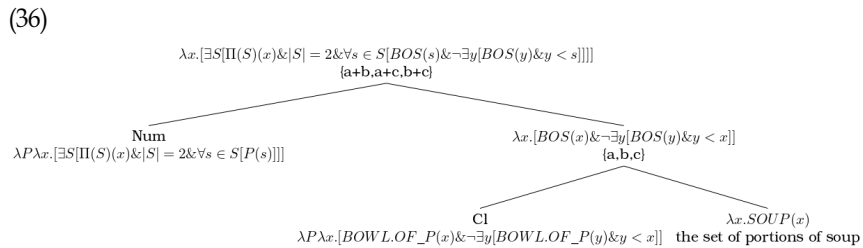
Furthermore, Deal proposes an atomizing function, AT, defined as follows (slightly simplified for exposition). Here, α introduces a variable over atomizing functions.

- (34) $[[an]] = \lambda P\lambda x.ATn(P)(x)$

We argue that the container classifiers (massifiers in the sense of Cheng and Sybesma, 1998) serve the atomizing function of Deal. We set aside here how to formally denote mass nouns and provide the following derivation.

- (35) a. liang wan tang
two CL.bowl soup
'two bowls of soup'
- b. $[[\text{tang}]] = \lambda x.\text{SOUP}(x) = \text{the set of all portions of soup}$
 $[[\text{wan}]] = \lambda P\lambda x.[\text{BOWL.OF_P}(x) \ \& \ \neg\exists y[\text{BOWL.OF_P}(y) \ \& \ y < x]]$
 $[[\text{liang}]] = \lambda P\lambda x.[\exists S[\Pi(S)(x) \ \& \ |S| = 2 \ \& \ \forall s \in S[P(s)]]]$

The following, then, is the derivation for (35a). Note that BOS is an abbreviation for the formal expression BOWL.OF.SOUP.⁷



In sum, we have shown that a semantic derivation along the lines of Little *et al.*, combined with Deal's analysis of count and mass nouns can provide a semantic analysis to support a uniform classifiers-for-nouns approach to Chinese classifiers. The analysis presented here requires refinement in future research; however, we note that there is a lack of consensus on the semantic treatment of mass nouns. See in particular footnote 6.

6. Conclusion

We have investigated Zhang's (2011, 2013) analysis of classifiers in Mandarin

⁷ We remain agnostic here as to whether the atomizing function is independent of the container classifier or is an integral part of it. An independent atomizer would compose with the noun followed by the container classifier. Evidence for a null atomizer that partitions mass substances out into portions comes from the fact that some languages contain overt singulatives (Mathieu 2012), which, as Deal explains, have nearly the exact same function as the atomizer.

Chinese along with Little *et al.*'s (2022) proposal on the two proposed structures of classifier constructions: classifiers-for-nouns and classifiers-for-numerals. Zhang proposes that Chinese has both types of structures; however, Little *et al.* argue that the two different structures have different semantics for numerals. Thus, if a single language had both types of classifiers constructions, it would have to list every numeral in the lexicon twice. Upon closer investigation of the arguments in Zhang, combined with the diagnostics from Little *et al.* we concluded that the facts in Chinese can be handled with only the classifiers-for-nouns model. Thus, only a single denotation is needed for numerals in Chinese. We ended with some brief suggestions on how the semantic derivation might proceed, leaving certain details to future work.

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